

Electron and Positron as Minimal Stable Field Modes of the UQSH

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Abstract

This paper presents a minimal formalization of the Universal Quantum Foam Hypothesis (UQSH) focusing on the nature of elementary states. Within this framework, the electron and positron are not interpreted as fundamental, isolated particles, but as minimal stable state modes of a dynamic field medium, the "Qu-foam". We introduce a regime-specific parameterization to describe the density of bound field organization and define the coupling threshold for detectable events. By introducing a discrete coupling parity, we account for the symmetry between matter and antimatter states without invoking additional substance-like properties. This model serves as an ontological interpretative layer that complements established quantum field theories by providing a field-mechanical cause for discrete stability classes and gravitational coupling.

1 Ontological Clarification

Within the UQSH, physical entities are not understood as fundamental particles, but as stable state modes of an underlying field medium. Designations such as “electron” or “matter” describe regime-specific, minimally stable organizational forms of bound field structure. Their apparent locality arises through threshold coupling to detectors and interactions, not through ontologically isolated substances.

2 Interpretative Positioning

The present work understands itself as an ontological interpretative layer within established physical theories. It neither claims to revise experimentally confirmed measurements nor to replace existing field equations.

In particular, the results of quantum mechanics and quantum electrodynamics remain unaffected. The model presented here aims to formulate a structural and ontological interpretation of discrete states such as the electron. Where existing theories provide precise predictions, these are not questioned but understood as effective regime descriptions. Within this framework, the UQSH does not claim to replace established theories, but to offer an interpretative extension.

3 Central Field Definition

The underlying medium of the Universal Quantum Foam Hypothesis is assumed to be fundamental. It is understood as dynamic, structure-forming, and nested in its manifestation. In the considered regime, the organizational state of this fundamental medium is parameterized by an effective scalar quantity:

$$Qu(x, t)$$

The quantity $Qu(x, t)$ does not constitute a complete microscopic description of the medium, but rather a regime-specific parameterization of its organizational structure. It serves as an order parameter describing local tension and structural states in the lower, weak regime.

4 Binding Density and Matter

The density of bound field organization ρ_{bind} and the condition for stable clusters:

$$\rho_{\text{bind}}(x) = \kappa \left(\frac{1}{2} K Q u(x)^2 + \frac{1}{2} L |\nabla Q u(x)|^2 \right)$$
$$\rho_{\text{bind}}(x) > \rho_{\text{crit}}(R)$$

The quantity $\rho_{\text{bind}}(x)$ measures locally stabilized field organization. The term $Q u(x)^2$ represents local tension binding, while $|\nabla Q u(x)|^2$ describes binding energy stored in spatial structure. The inequality $\rho_{\text{bind}}(x) > \rho_{\text{crit}}(R)$ defines the regime-dependent coupling threshold above which a bound state appears as a detectable event.

5 Dynamics and Evolution

The temporal change of the field with source term J :

$$\partial_t Q u = F(Q u, \nabla Q u, \dots) + J(x, t)$$

The function F describes the intrinsic relaxation dynamics of the medium. The source term $J(x, t)$ represents physical interventions or couplings to other systems. An explicit form of F is not specified in this minimal model.

6 Energy (Ontological Interpretation)

The energy of a stable state mode is understood in the minimal model as integrated bound field organization:

$$E = \int \rho_{\text{bind}}(x) d^3x$$

Energy is thus not interpreted as an independent substance-like quantity, but as a measure of stabilized field organization. A numerical derivation of experimental parameters (e.g., electron mass) is not claimed here.

7 Electron and Positron as Coupling Parities

To distinguish two regime-specific yet otherwise identical minimal modes, a discrete coupling parity is introduced:

$$\sigma \in \{+1, -1\}$$

Here, $\sigma = +1$ corresponds to the electron mode and $\sigma = -1$ to the positron mode in regime R . These two states are understood as expressions of a symmetric organizational structure of the medium. Detector coupling is not modeled as an intrinsic particle property, but as a sign structure of coupling to bound field organization:

$$C_R[Qu; \sigma](x) = \sigma g_R$$

A detectable event occurs when the coupling exceeds a regime-dependent threshold:

$$|C_R[Qu; \sigma](x)| > C_{\text{crit}}(R)$$

Thus, the minimality/stability of the mode remains independent of the sign, while the interaction direction is distinguished by σ .

8 Conservation Structure

In the case of pair creation or annihilation, a structural conservation of coupling parity holds in the minimal model:

$$\sum \sigma = 0$$

This relation is understood as an expression of a symmetric organizational structure. In the minimal model, local pair formation is assumed to be complementary. A possible global asymmetry of coupling parity is not excluded and would be understood as an expression of a weak structural preference of the medium.

9 Spin (Ontological Interpretation)

Spin is not understood as a classical rotation of a material object, but as an intrinsic orientational or chirality structure of the minimally stable mode. Its

quantization is interpreted as the expression of a discrete stability class. A derivation of the spin-1/2 structure or of the magnetic moment is not claimed in this minimal model.

10 Gravitational Coupling

The effective potential and the resulting acceleration:

$$\Phi(x) = \alpha Qu(x)$$

$$\vec{a}(x) = -\alpha \nabla Qu(x)$$

$$\nabla^2 Qu(x) \propto \rho_{\text{bind}}(x)$$

The identification $\Phi(x) = \alpha Qu(x)$ represents an effective regime interface and is to be understood as an approximation in the quasi-static, weak-field regime. The Poisson-like relation $\nabla^2 Qu \propto \rho_{\text{bind}}$ connects bound field organization with effective gravitation.

11 Scaling

The dimensionless scale indicator for the transition L^* :

$$\varepsilon(L) = \frac{L|\nabla Qu|}{|Qu|}$$

$$\varepsilon(L^*) \approx 1$$

The dimensionless parameter $\varepsilon(L)$ characterizes the transition between homogeneous and structure-dominated behavior. The scale L^* is a regime interface scale defined by $\varepsilon(L^*) \approx 1$, but not a fundamentally postulated minimal or maximal length.

12 Open Points (Remarks)

This minimal model deliberately leaves several aspects open:

- Concrete specification of the dynamical function F .
- Determination of the parameters κ, K, L, α and the thresholds $\rho_{\text{crit}}(R), C_{\text{crit}}(R)$.

- Precise connection to full quantization (without claiming to replace established QFT).

The work understands itself as a structural and ontological framework, not as a complete replacement theory.

13 Falsifiability

The UQSH would be falsified if elementary states could be described entirely without field-like organizational structure, or if stable discrete states could not be assigned to structural stability classes. In particular, an experimentally confirmed description of elementary particles as ontologically isolated substances without underlying field organization would contradict the framework presented here.

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